

Colorectal cancer, also known as CRC, is one of the most commonly diagnosed cancers in men and women. This disease develops when abnormal cells grow uncontrollably in the colon and/or rectum. These cells initially begin as small polyps but may become cancerous over time. CRC is preventable and treatable if detected early enough through screenings. Screenings allow for the removal of polyps and can diagnose cancer at early stages, when treatment is more likely to have positive outcomes. Despite this, CRC screening rates remain below national goals, especially among less fortunate demographics.

Screening for CRC is recommended starting at around age 45 for patients at average risk. If the patient's family has a history of CRC, then an earlier age is recommended. CRC screenings should be incorporated into men's and women's preventive health routines as they age. Screening methods range from colonoscopy procedures—which some patients may feel are too invasive—to less invasive stool tests. Colonoscopies are considered the best available test because they enable the detection and removal of polyps. Stool tests are more accessible and convenient but may still require a follow-up colonoscopy depending on initial results.

Despite the effectiveness of CRC screenings, the number of people who undergo them remains low. Patients report that limited knowledge about guidelines, concerns about discomfort, shame and embarrassment, fear of results, and logistical difficulties discourage them from pursuing CRC screenings. Other barriers include rigid work schedules, lack of insurance coverage, limited access to screenings, and emotional resistance.

Primary care “safety-net” systems that serve low-income and medically underserved populations face unique barriers as well. Competing demands during short visits, lack of reminder systems, and limited capacity for follow-up appointments hinder the delivery of CRC

screenings. Preventive care is often not the highest priority when patients present with urgent health or social needs. Providers may also struggle due to limited access to culturally appropriate educational and informational resources.

CRC also has a familial component; patients with a family history of CRC or hereditary cancers are at significantly higher risk than those without. This raises issues of transparency within families. Some families are open about hereditary cancers, which can increase screening rates, while others are dismissive and downplay the risk. These differences often reflect interpersonal and cultural dynamics. Stigma also consistently emerges as a barrier to CRC screenings. Patients may feel embarrassed or uncomfortable about the colonoscopy process, which requires direct access to the rectum. They may also fear being judged about their bowel habits or family history of cancer. Cultural taboos further reinforce these barriers.

Opportunities for design in CRC treatment lie primarily in the media surrounding CRC. The creation of advertisements and infographics that normalize CRC and humanize those affected by it may encourage more people to pursue screenings and treatment. While designing media that is effective across all demographics would be ideal, a one-size-fits-all approach is unlikely to succeed. Therefore, it is important to design targeted media for specific demographics to maximize effectiveness.

CRC remains a major public health concern, but it is effectively treatable and preventable when identified early. The biggest issue with CRC screenings is not their effectiveness or reliability, but rather their accessibility and appeal across all demographics. Overcoming these barriers is essential, as doing so will immediately improve the health and outcomes of those at risk.